



**SCHOOL OF COMPUTER SCIENCES UNIVERSITI SAINS MALAYSIA CMT221: Database
Organization and Design**

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System Design

[Group Number 42]

Dr. Azleena Mohd Kassim & Dr. Hazrina Yusof Hamdani

Project Title: Greening Public Spaces For Sustainable Communities

Name	Matric No	USM Email Address	role	major
AHMED M.H. Taha	22304588	ahmedtaha@student.usm.my	LEADER	IC
ABDELRAHMAN HAMDY	160703	abdelrahman@student.usm.my	MEMBER	IC
ABDULLAH ABDULRAHMAN	22304585	abdullahalbaker@student.usm.my	MEMBER	CI
LI BINGYE	22304752	bingye@student.usm.my	MEMBER	IC

Table of Contents

1. Introduction	3
2/3 User requirements and the Business Rules.....	4
2/3-1 User Registration Module:	4
2/3-2 Posts Management Module:	5
2/3-3 Feedback Management Module:	7
2/3-4 Events Management Module:	8
4. Entity relationship modelling.....	9
4-1 Users Registration Module :	9
4-2 Posts Management Module:	11
4-3 Feedback Management Module.....	14
4-4 Events Management Module.....	16
4-5 Combined-Erd	18
5. Overall format and documentation	20

1. Introduction

This project is about developing a database system for an organization called **Greening Public Spaces for Sustainable Communities**. The organization focuses on improving urban spaces by addressing issues like plastic waste, smoking-related litter, and protecting the ozone layer. Its goal is to encourage people to live more sustainably and help create better cities, in line with **Sustainable Development Goal (SDG) 11**, which aims to make cities more sustainable and healthier.

Currently, the organization faces challenges such as managing public space clean-ups, sharing information efficiently, and tracking community involvement. These problems make it difficult for the organization to reach more people and promote environmental action effectively.

The new database system will help by providing a platform where people can register, join events, provide feedback, and engage with posts. The system will also feature tools to advertise events and raise awareness about the organization's work. To make the organization more widely known, the system will include **advertising tools**, allowing the organization to post in groups, share in newspapers, and promote through other platforms. This will help attract a larger audience, encouraging more people to get involved in sustainability efforts.

The system will have several modules:

User Module: Allows users to sign up, manage their accounts, and participate in events and discussions.

Posts Module: Lets the organization post information about environmental issues like pollution and ozone protection.

Feedback Module: Enables users to provide feedback on events, helping the organization improve future activities and engage with the community.

Event Module: allows the organization to manage community events by enabling users to register, view event details, track participation, while also allowing administrators to create, edit, and promote events to raise awareness and encourage involvement in sustainable actions.

This new system will make it easier for the organization to manage events, communicate with the public, and promote a greener, more sustainable future for public spaces while increasing visibility through strategic advertising

2/3 User requirements and the Business Rules

2/3-1 User Registration Module:

Table1: User Registration Module User Requirements and Business Rules

	User Requirements	Business Rules
User and Contact Relationship	A user must be able to register on the platform by providing their name, email address, a secure password, and selecting a specific role (e.g., Regular User or Administrator). Registration is required to access system features such as commenting, subscribing to newsletters, or contacting administrators.	A user sends one or many contacts . Each contact is sent by one and only one user .
User and Role Relationship	The user must have the ability to update their profile details, such as changing their password, email address, or name. Profile management should be simple, secure, and only accessible to the authenticated user.	A user has one and only one role . Each role belongs to one or many users .

2/3-2 Posts Management Module:

Table2: Posts Management Module User Requirements and Business Rules

	User Requirements	Business Rules
User and Posts	The system must allow users to create and manage posts. Each user can create zero or many posts, and each post must be created by one and only one user. Posts will include details such as a title, status, content, the associated user who created it, the category it belongs to, and timestamps for creation and deletion. Users must be able to view, edit, and delete their own posts but must not be allowed to modify posts created by other users.	A User can create zero or many Posts, each Post is created by one and only one User.
Post and Comments	The system must allow posts to have comments. A post can have zero or many comments, and each comment must belong to one and only one post. Comments will include the content, the user who wrote them, the post they are associated with, and timestamps for creation and deletion. Users must be able to view comments associated with a post, add their own comments, and delete only the comments they have written. Comments written by other users must not be modifiable by anyone except the original author.	A Post has zero or many Comments, each Comment belongs to one and only one Post.
User and Comments	The Comment section lets users add, view, and delete comments on posts. Each comment includes a unique ID, content, user and post IDs, a creation date, and a delete flag (Comment-Delete) . Users can manage their own comments, while admins can moderate all comments, including deleted ones. The system ensures valid data, proper links to posts and users, and secure access control.	A User can writes zero or many Comments, each Comment is written by one and only one User.
Post and Post Categories	Each post must belong to one and only one category, and a category can have zero or many posts. Categories will include attributes such as the category name, description, and the user who created them. Users must assign a category when creating a post. The system should ensure that a post cannot exist without a category and must allow users to view, edit, or delete the categories they have created. Categories created by one user must not be modifiable by other users.	A Post must have one or many Post_Category, and a Post_Category may had zero or many Posts.

	User Requirements	Business Rules
User and Post Categories	Users can had zero or many post categories, and each category must be belong to one and only one user. A user must be able to view a list of categories they have created, edit or delete their own categories, and assign these categories to their posts. Additionally, the system must allow users to read the categories they have authored and ensure that no unauthorized user can modify categories they do not own.	A User had zero or many Post Categories , and each Post_Category belongs to one and only User .

2/3-3 Feedback Management Module:

Table3: Feedback Management Module User Requirements and Business Rules

	User Requirements	Business Rules
User and Feedback_Users	The system must allow users to submit feedback tied to their identity. Each user can provide zero or many feedback entries, and each feedback must be linked to one and only one user. Feedback entries include attributes such as content, rating, and timestamps. Users must be able to view, edit, and delete their own feedback. Administrators can generate reports to track feedback submissions by users.	A user can submit one or many pieces of feedback . Each feedback is submitted by one or many users .
Feedback and Feedback_Users	The system must allow feedback entries to involve one or multiple users, supporting collaborative input. Each feedback entry includes content, rating, and timestamps, while tracking the users associated with it. Users must be able to view the feedback they contributed to, and administrators must be able to monitor and identify all users linked to feedback.	A feedback is written about one and only one event. Each event can receive one or many feedback entries.
Feedback and Feedback_Events	The system must allow users to submit feedback specific to events they attended. Each feedback must belong to one and only one event, while events can have zero or many feedback entries. Feedback attributes include content, numeric ratings, and timestamps. Users can browse feedback from other participants for events they attended, and administrators can generate reports summarizing feedback for event evaluation.	An event can be reviewed by one or many feedback entries. Each feedback reviews one and only one event.
Feedback_Events and Feedback_Users	The system must connect feedback entries to both the users who submitted them and the events they evaluate. Each feedback is tied to a single event and user. Feedback attributes include content, rating, and timestamps, providing detailed links between users and events. Administrators must be able to filter feedback by user or event for analysis and reporting.	A user can participate in one or many pieces of feedback. Each feedback can involve one or many users.

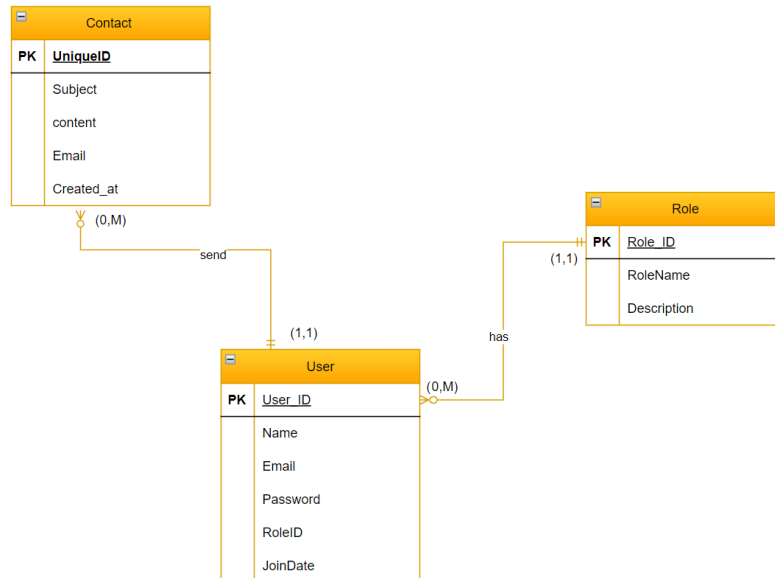
2/3-4 Events Management Module:

Table4: Events Management Module User Requirements and Business Rules

	User Requirements	Business Rules
User and Event Users (Registrations)	The database must track Event registrations, enabling Users to register for Events, view the Events they are registered for, and cancel registrations if needed. Event creators must be able to view and manage the list of registered participants for their Events. The system should enforce participant limits, restrict duplicate registrations, and provide tools for monitoring and updating registrations effectively.	An Event User is associated with one and only one User . Each User is associated with one or many Event Users .
User and Event Category	The database must allow Users to create and manage Event Categories by specifying a name and description. Users should associate Events with appropriate Categories during creation and be able to browse a list of available Categories. Administrators should have the ability to review and manage all Categories, while the system supports filtering and searching for Events by Category.	An Event Category is created by one and only one User . Each User creates one or many Event Categories .
Event and Event Users (Registrations)	The database must maintain a link between Events and their registered Users, enabling Event creators to monitor participant details and download attendee lists. Users should be able to view the Events they are registered for and manage their participation, while the system ensures accurate tracking of registrations and enforces capacity limits to prevent overbooking.	An Event User is associated with one and only one Event . Each Event is associated with one or many Event Users .
Event and Event Category	The database must categorize Events under specific Event Categories to improve organization and searchability. The system should provide summaries for each Category, including the number of Events and total participants, ensuring efficient navigation and management of Event data.	An Event is categorized under one and only one Event Category . Each Event Category categorizes one or many Events .

4. Entity relationship modelling

4-1 Users Registration Module :



Users Registration Module

Diagram 1 : The Diagram above show is the crow foot notation for the Events Management Module

User:

The Users entity represents the core participants of the system. Each user is identified by attributes such as User_ID, Name, Email, Password, RoleID, and JoinDate. Users play a key role in the system by sending contact messages, being assigned specific roles that determine their access levels, and interacting with various components such as events, posts, and feedback.

Contact:

Represents the messages or inquiries submitted by users. It contains attributes such as UniqueID, Subject, Content, Email, and Created_at. This entity allows users to communicate with administrators or the system, ensuring that user concerns and feedback are appropriately stored and addressed.

Role:

Represents the roles assigned to users. It contains attributes such as Role_ID, RoleName, and Description. Roles are essential for defining user permissions and restricting access to certain functionalities based on their responsibilities.

Relationships

User to Contact:

The **User to Contact** relationship ensures that users can send multiple contact messages, but each message is associated with only one user. This establishes clear accountability for each contact while allowing users to communicate with the system freely.

Cardinality: One-to-Many

Explanation: A user can send multiple contact messages, and each contact message must belong to exactly one user.

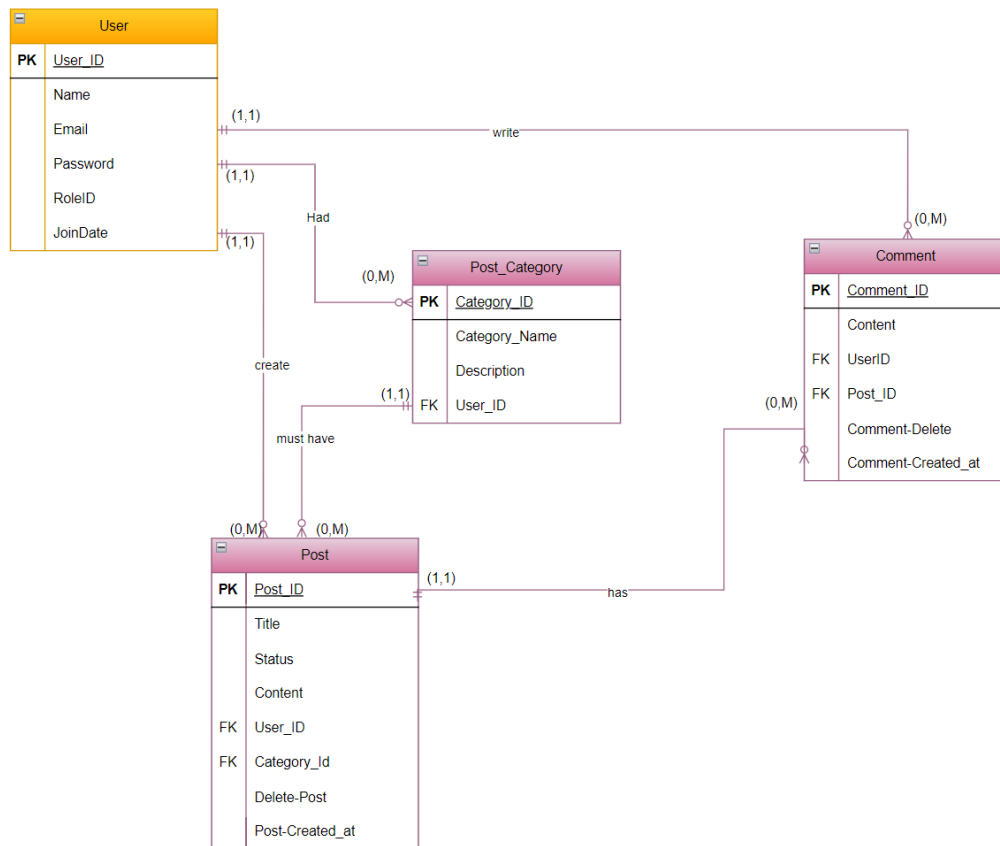
User to Role:

The **User to Role** relationship links users with their designated roles within the system. Each user is assigned one role, such as "Admin" or "Member," while a role can be assigned to multiple users. This structure supports role-based access control, ensuring that users have permissions appropriate to their responsibilities.

Cardinality: One-to-Many

Explanation: A user can have only one role, but a role can be shared by multiple users. For example, multiple users might share the "Member" role, granting them similar levels of access.

4-2 Posts Management Module:



Posts Management Module

Diagram2 : The Diagram above show is the crow foot notation for the Events Management Module

Entities

Post: Represents the posts created by users. It contains attributes such as Post_ID, Title, Status, Content, timestamps (Post-Created_at and Delete-Post), and foreign keys to associate it with a User_ID and a Category_ID. Posts are the core content of the module.

Comment: Represents the comments users write on posts. It contains attributes like Comment_ID, Content, Comment-Created_at, and Comment-Delete, along with foreign keys linking it to UserID and Post_ID. Comments facilitate user interaction and discussion.

Post_Category: Represents the categories to which posts belong. It contains attributes such as Category_ID, Category_Name, Description, and along with foreign key which is User_ID to track the creator. Categories help organize posts for better accessibility and management.

Relationships

User to Post

The User to Post relationship ensures that each post is created and managed by a single user, while allowing users to create multiple posts. This establishes clear ownership and accountability for each piece of content, supporting a personalized content creation experience.

User to Comment

The User to Comment relationship highlights how users can contribute to discussions by writing comments on posts. It ensures that every comment is linked to one user, facilitating accountability while allowing for multiple comments per user on different posts.

Post to Comment

In the Post to Comment relationship, posts serve as the base for user interaction through comments. Each post can have many comments, allowing for dynamic discussions, but every comment must be linked to a single post, ensuring that feedback is directly related to the content.

Post to Post_Category

The Post to Post_Category relationship ensures that every post belongs to a category, helping to organize content. Categories can have multiple posts, allowing for efficient content sorting and easier discovery by users.

User to Post_Category

Finally, the User to Post_Category relationship enables users to have and manage categories, offering them the ability to organize posts in a way that fits their preferences. This flexibility enhances the user experience by ensuring that categories are linked to their creators and can be easily edited or deleted.

Cardinality of the Relationships

User to Post

Cardinality: One-to-Many

Explanation: This ensures that posts are associated with their creators, providing ownership and accountability.

User to Comment

Cardinality: One-to-Many

Explanation: This allows users to engage in discussions by contributing multiple comments on various posts.

Post to Comment

Cardinality: One-to-Many

Explanation: This supports user interaction by enabling multiple comments per post, fostering discussion.

Post to Post_Category

Cardinality: One-to-Many

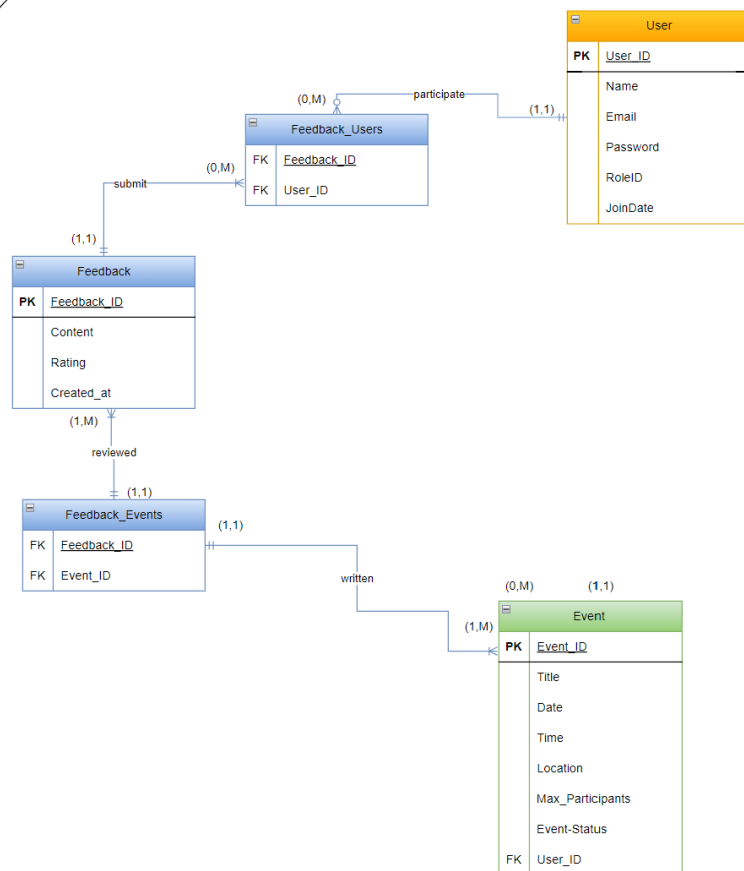
Explanation: Posts must be categorized for better organization, but categories can include multiple posts.

User to Post_Category

Cardinality: One-to-Many

Explanation: This allows users to manage their own categories independently, ensuring flexibility in organization.

4-3 Feedback Management Module



FEEDBACK Management Module

Diagram3 : The Diagram above show is the crow foot notation for the Events Management Module

Entities:

Feedback: The feedback given by users regarding events. It contains content, a rating, and a timestamp. Feedback is crucial for evaluating events and gaining insights into user experiences.

Feedback_Users: An associative entity linking users to the feedback they provide. This is crucial for tracking which users contributed to which pieces of feedback.

Feedback_Events: Another associative entity that connects feedback to events, ensuring each piece of feedback is associated with a specific event.

Relationships:

User to Feedback_Users: A user can submit feedback for one or more events. The relationship is many-to-many, as each feedback can involve multiple users. This gives flexibility for group or collaborative feedback.

Feedback to Feedback_Users: A feedback entry can be tied to one or many users, making this a many-to-many relationship. This allows for the scenario where multiple users can provide feedback for the same event.

Feedback to Feedback_Events: Feedback is linked to an event it evaluates. Each feedback is tied to one event, and each event can have multiple feedback entries. This relationship ensures that the feedback is event-specific, allowing the system to evaluate each event based on user input.

Feedback_Events to Feedback_Users: This is a many-to-many relationship where feedback entries are linked to both users and events. This allows for the proper tracking of feedback from each user for a particular event.

Cardinality of the Relationships:

User to Feedback_Users (One-to-Many)

Cardinality: One-to-Many

Explanation: This allows users to contribute to multiple feedback entries, and multiple users can be associated with a single feedback entry.

Feedback to Feedback_Users (One-to-Many)

Cardinality: One-to-Many

Explanation: The feedback entries can involve one or more users, and each user can write feedback for multiple events.

Feedback to Feedback_Events (One-to-Many)

Cardinality: One-to-Many

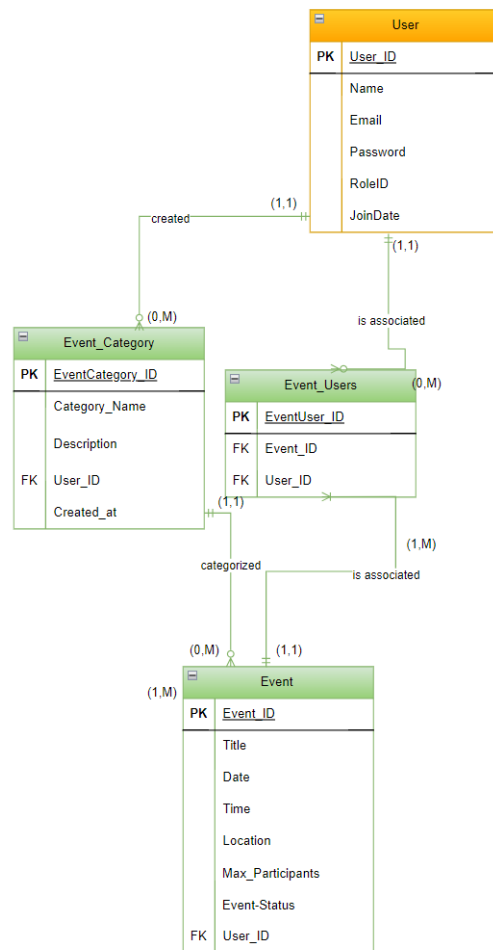
Explanation: Each feedback entry is specifically for one event, but an event can have multiple feedback entries associated with it.

Feedback_Events to Feedback_Users (One-to-Many)

Cardinality: One-to-Many

Explanation: This relationship allows multiple users to provide feedback on the same event, and each user can give feedback on many events.

4-4 Events Management Module



Events Management Module

Diagram4 : The Diagram above show is the crow foot notation for the Events Management Module

Event: This entity stores information about community events, including details such as Event_ID, Title, Date, Time, Location, Max_Participants, and Event_Status. The Event_ID is the primary key. Events are tied to specific users (event organizers) and categories, linking them to users' sustainability efforts.

Event Category: This entity categorizes events, making it easier for users to search and filter events. Each category has attributes like EventCategory_ID, Category_Name, Description, and Created_at, with EventCategory_ID as the primary key. Categories help organize events, such as "Clean-Up Drives" or "Workshops."

Event_Users: This relationship table tracks users' participation in events. It includes the EventUser_ID as the primary key and foreign keys (Event_ID and User_ID) to establish which users are participating in which events. This table captures the many-to-many relationship between users and events.

User and Event Users (Registrations): Users can register for many events, and an event can have multiple participants, so this relationship is also many-to-many (M:M), facilitated by the Event_Users table. This relationship enables the system to manage user registrations for events and track participants.

User and Event Category: A user can create multiple event categories, but each event category is associated with only one user who is responsible for it. This relationship is one-to-many (1:M), with the user acting as the creator of categories that help organize events.

Event and Event Category: Each event belongs to one category, but a category can contain many events. This is a one-to-many (1:M) relationship, which allows events to be grouped into relevant categories, improving discoverability and user experience when browsing or searching for events.

Event and Event Users (Registrations): Each event can have many users registered, and each user can be registered for multiple events, making this a many-to-many (M:M) relationship, again managed by the Event_Users table. This relationship ensures the system can track who is participating in which events and manage registration data efficiently.

4-5 Combined-Erd

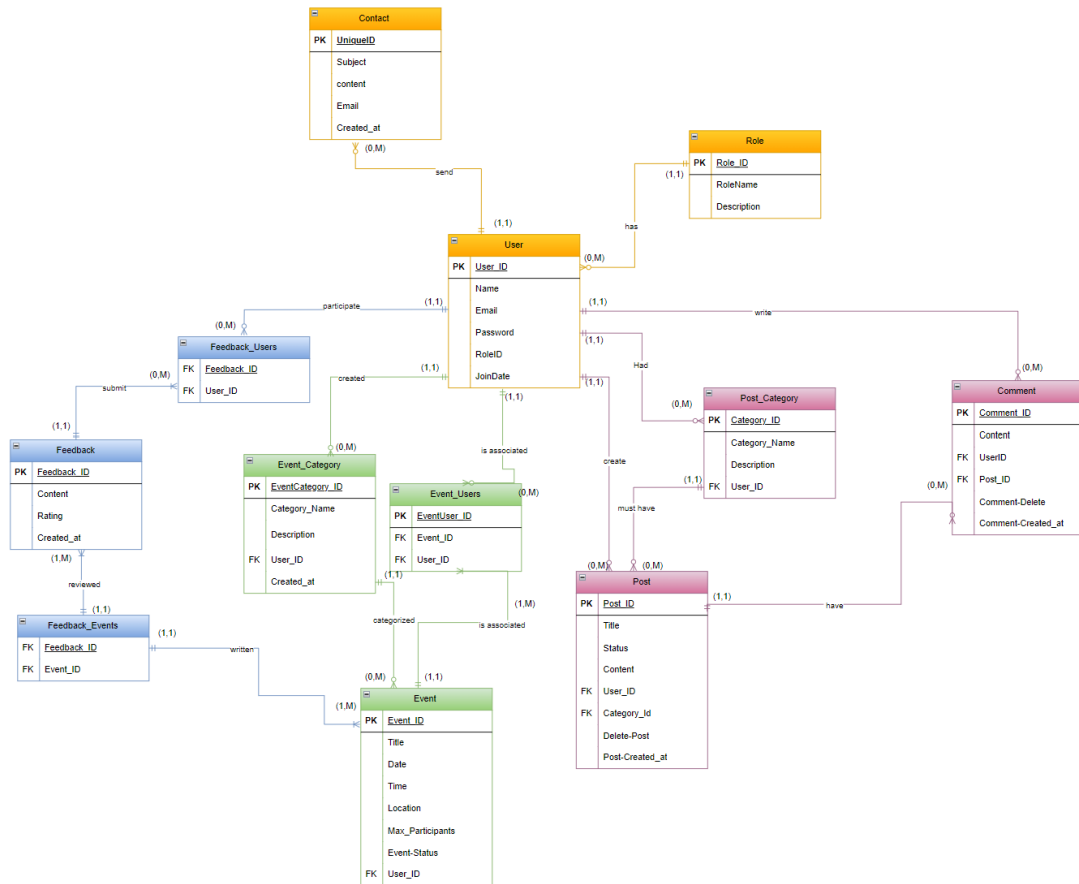


Diagram5 : The diagram above represents the combined Crow's Foot notation for all modules.

There are 12 entities in the provided ERD are:

1. Contact
2. User
3. Newspaper
4. Post_Category
5. Post
6. Comment
7. Feedback
8. Feedback_Users
9. Event_Category
10. Event
11. Event_Users

12. Feedback_Events

Conclusion:

This ERD model captures the complex interactions between users, posts, feedback, and events. It covers a broad range of functionalities, including content creation, feedback collection, and event participation. The relationships and cardinalities ensure that the system can handle multiple users interacting with multiple posts, and events, all while maintaining a clear structure for tracking interactions, feedback, and event management.

5. Overall format and documentation

The documentation adheres to the required formatting guidelines, ensuring professionalism and compliance with institutional standards. The content is written in Times New Roman font, size 11, with 1.5-line spacing for readability. All text is justified, aligning it to both left and right margins, and formatted to A4 page size (8.27" x 11.69"). Each page is properly numbered to facilitate easy navigation.

The report includes a cover page, a mandatory requirement for all submissions, including project proposals, system design, and system implementation. Reports without a cover page will not be marked, and the format adheres to the sample provided in Section X. A table of contents is also included for system design and system implementation reports, providing an organized overview of the document's structure.

The content is structured into several key sections. The introduction provides an overview of the project, outlining its objectives, scope, and relevance to sustainable development goals. The requirements and rules section elaborates on the user requirements and business rules for each module, presented in a clear, tabular format. The entity relationship modelling section details the database structure, including entities, attributes, and relationships, with diagrams for better understanding. The implementation details describe how different modules were designed and integrated, while the testing and validation section summarizes methodologies, testing scenarios, and results to validate system functionality. The report concludes with a conclusion and recommendations, reflecting on the project's achievements and suggesting areas for improvement.

Figures and tables are properly labeled and cross-referenced in the text to ensure clarity and usability. The database schema, including primary and foreign keys, is explained alongside ER diagrams to provide a comprehensive view of the system design. A user manual is included to guide users on system features, and a testing report summarizes the results of user acceptance testing, bug fixes, and optimizations. Additionally, the report includes a references and bibliography section to ensure proper citation of all external sources used.

Version control is maintained with a revision history documenting all updates to the design and functionality. A changelog highlights major improvements, while an acknowledgment section recognizes contributions from team members, supervisors, and supporters. Submission details, including the group number, team member names, and project title, are featured on the cover page to ensure compliance with submission guidelines. This format ensures that the report is comprehensive, well-organized, and adheres to all specified requirements.

Reference:

Coronel, C. and Morris, S., 2017. Database Systems: Design, Implementation, and Management. 12th ed. Boston: Cengage Learning. ISBN 978-1-305-62748-2.